DO NOW — RECALL

Thinking ahead: inputs, caching and reuse

Name: _____ Class: _____ Date: _____

1. Why omit cover colour from a simple book-loan model?

2. What is a reusable subroutine?

3. Calculate 7×7.

4. Why is caching squares safer than caching changing attendance data?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Cover colour does not determine availability.

2. Model answer

A function/procedure with a clear interface usable in multiple places.

3. Model answer

49.

4. Model answer

Squares of the same integer do not change; attendance does.

Vocabulary — know these terms

input — Data passed into a computation.

output — Result of a computation.

precondition — Condition required before a process can run correctly.

cache hit — Requested value found in storage.

cache invalidation — Removing or refreshing outdated stored results.

NEW LEARNING

Read, annotate and explain

Define the contract

The input is an integer, and the output is its square; overflow is a possible precondition concern.

Trace repeated work

First call for 5 calculates and caches 25; second returns the stored result.

Evaluate caching

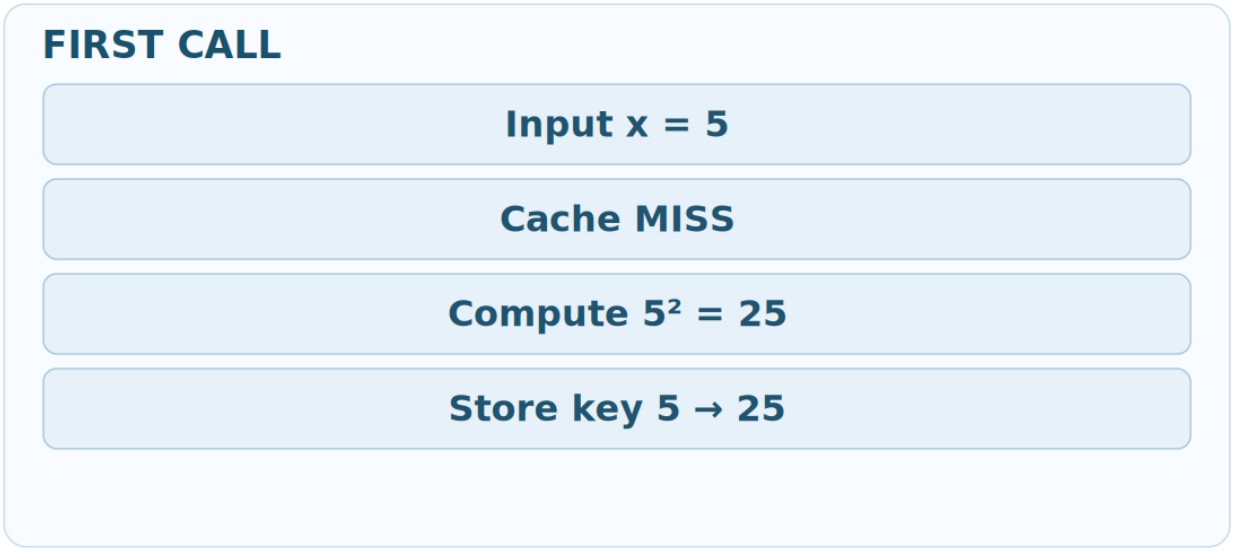
Saves repeated calculation but costs memory. For changing data, cached values can become out of date.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

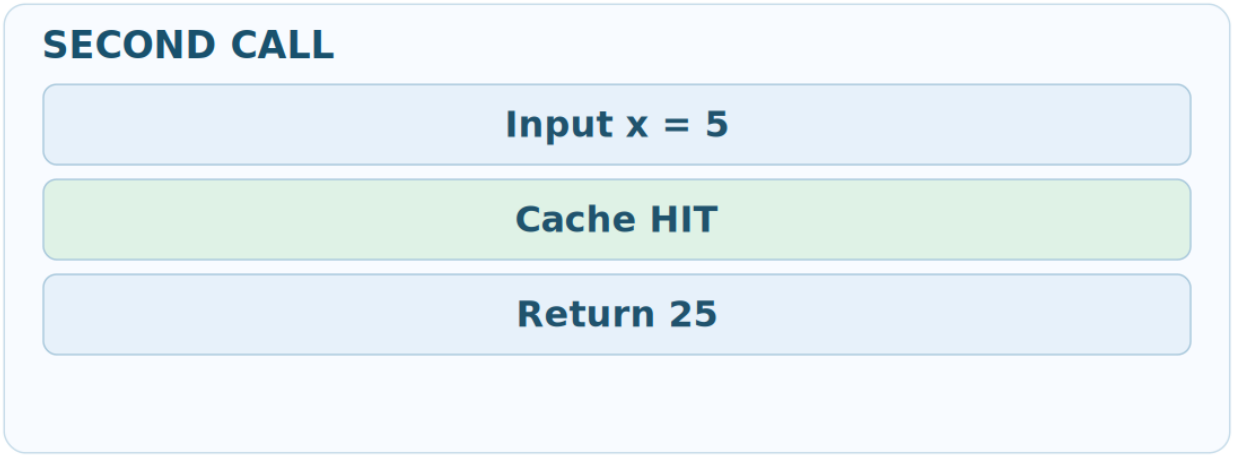
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



FIRST CALL; Input x = 5; Cache MISS; Compute $5^2 = 25$; Store key 5 → 25

New Learning



SECOND CALL; Input x = 5; Cache HIT; Return 25

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



STALE DATA RISK; Live value changes; Old cache remains; Invalidate / refresh

I DO — WORKED EXAMPLE

Follow the worked method

A cached ticket price for zone A is £3.40. The same zone is requested twice.

Predict first

The key A is already stored, so the lookup uses the cache.

Model and check

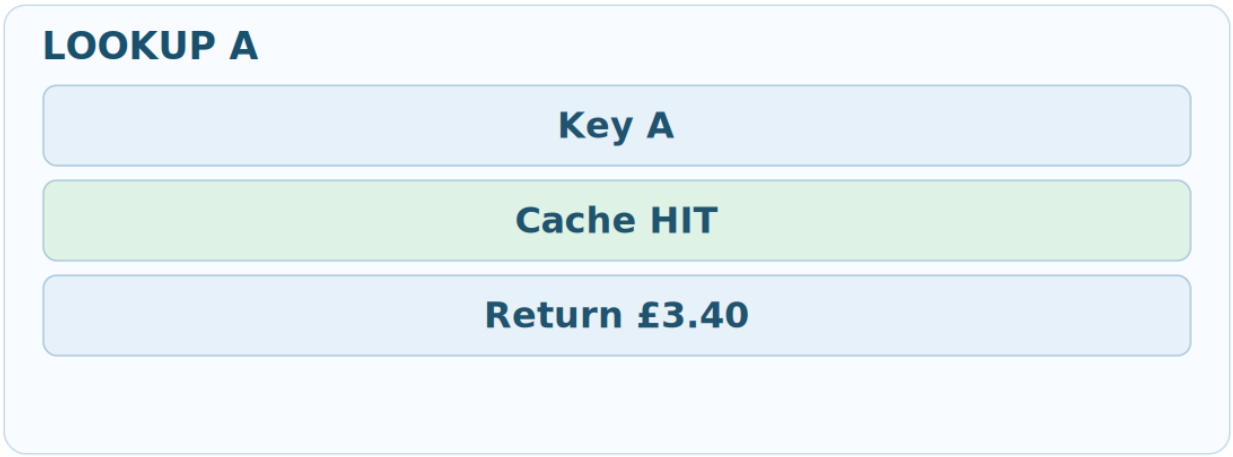
If pricing changes, the cached price must be updated or expired.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A cached ticket price for zone A is £3.40. The same zone is requested twice.

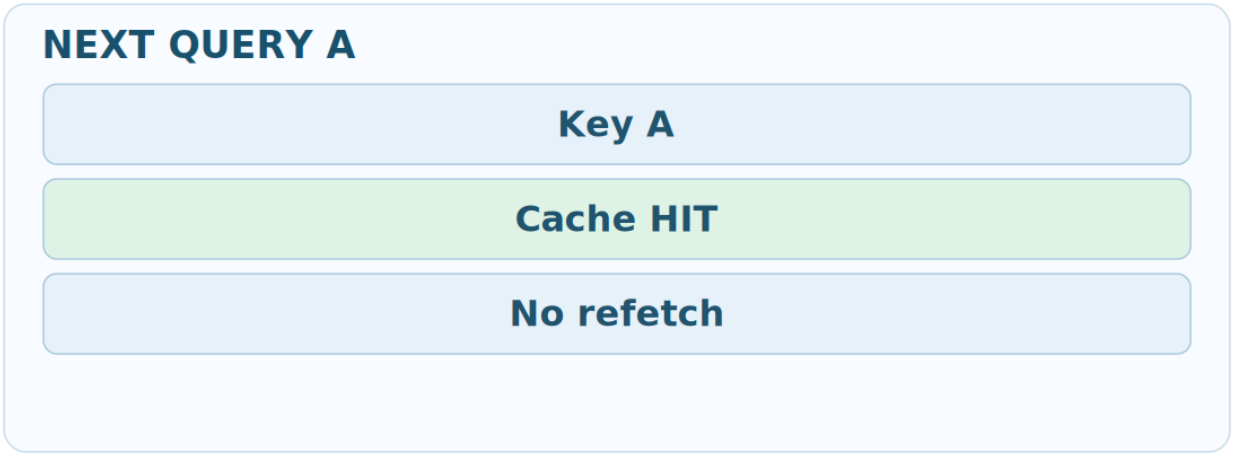
The key A is already stored, so the lookup uses the cache.



LOOKUP A; Key A; Cache HIT; Return £3.40

A cached ticket price for zone A is £3.40. The same zone is requested twice.

The key A is already stored, so the lookup uses the cache.



NEXT QUERY A; Key A; Cache HIT; No refetch

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A cached attendance total is 12 but the live total changes to 15. How should the system respond?

Using the old cached 12 gives a stale answer instead of the new total 15.



STALE CACHE; Cached 12; Live 15; Wrong answer

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first Using the old cached 12 gives a stale answer instead of the new total 15.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

For SquareCached(6), identify input, output and two preconditions or assumptions.

Question 2

Trace two calls of SquareCached(7) starting with an empty cache.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Give three advantages/disadvantages of caching changing school data and a way to avoid stale values.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Once a cached value is stored it is always correct.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

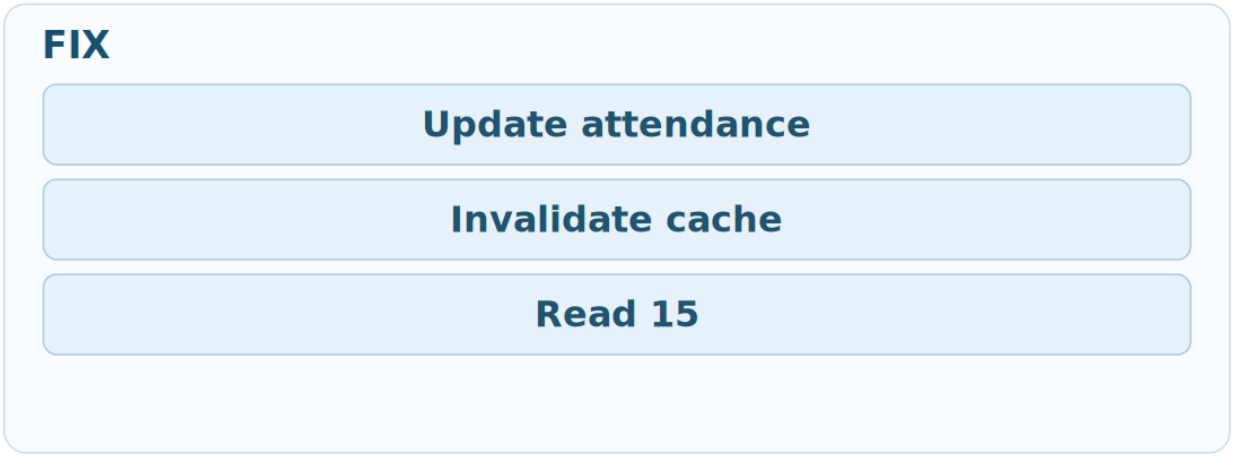
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A cached attendance total is 12 but the live total changes to 15. How should the system respond?

Using the old cached 12 gives a stale answer instead of the new total 15.



FIX; Update attendance; Invalidate cache; Read 15

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model and check

Update or invalidate the cached value when attendance changes, or apply a suitable expiry policy.

IMPROVE: Explain one correction to your own reasoning.